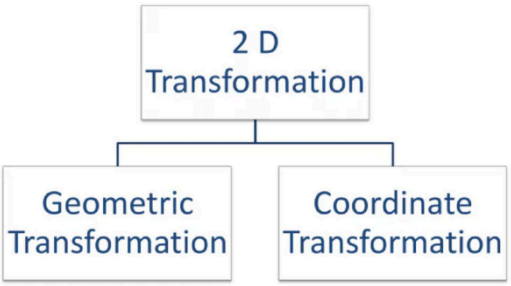
Unit 3

2D & 3D Transformation

2D Transformation: Transformation means changing some graphics into something else by applying rules. We can have various types of transformations such as translation, scaling up or down, rotation, shearing, etc. When a transformation takes place on a 2D plane, it is called 2D transformation.



There are two complementary points of view for describing object transformation.

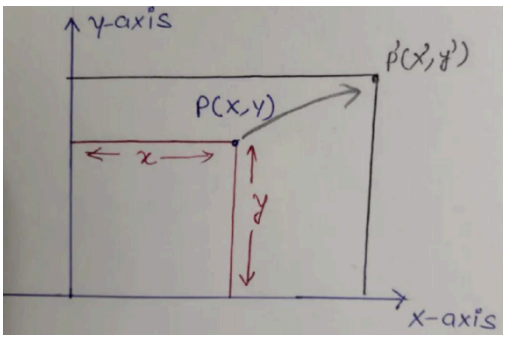
1. Geometric Transformation: The object itself is transformed relative to the coordinate system or background. The mathematical statement of this viewpoint is defined by geometric transformations applied to each point of the object.
2. Coordinate Transformation: The object is held stationary while the coordinate system is transformed relative to the object. This effect is attained through the application of coordinate transformations.

Homogeneous Coordinates: To shorten this process, we have to use a 3×3 transformation matrix instead of a 2×2 transformation matrix. To convert a 2×2 matrix to 3×3 matrix, we have to add an extra dummy coordinate W. In this way, we can represent the point by 3 numbers instead of 2 numbers, which is called Homogeneous Coordinate system.

Types of Transformations:

1. Translation
2. Scaling
3. Rotating
4. Reflection
5. Shearing

Translation: It is the straight line movement of an object from one position to another is called Translation. Here the object is positioned from one coordinate location to another.



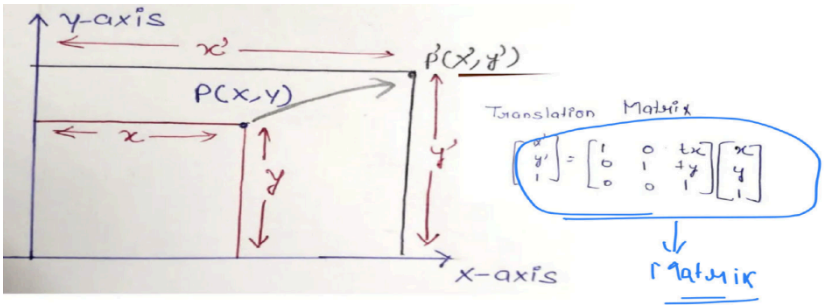
To translate a point from coordinate position (x, y) to another (x1 y1), we add algebraically the translation distances Tx and Ty to original coordinate.

$$\mathbf{x'=x+t_x}$$

$$\mathbf{y'=y+t_y}$$

The translation pair (t_x,t_y) is called as shift vector.

Translation is a movement of objects without deformation. Every position or point is translated by the same amount. When the straight line is translated, then it will be drawn using endpoints.



Translation Matrix

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Que To translate a traingle with vertices at original co-ordinates (10, 20), (10, 10), (20, 10) by $t_x = 5$, $t_y = 10$

Short tricks

Sol^n

$$\begin{bmatrix} 10+5 = 15 \\ 10+5 = 15 \\ 20+5 = 25 \end{bmatrix} \quad \begin{bmatrix} 20+10 = 30 \\ 10+10 = 20 \\ 10+10 = 30 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 10 & 20 \\ 20 & 10 & 10 \\ 1 & 1 & 1 \end{bmatrix}$$

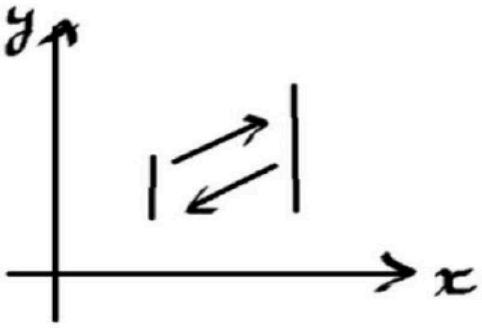
$$= \begin{bmatrix} 15 & 15 & 25 \\ 30 & 20 & 20 \\ 1 & 1 & 1 \end{bmatrix}$$ Ans

Scaling: It is used to alter or change the size of objects. The change is done using scaling factors. There are two scaling factors, i.e. S_x in x direction S_y in y-direction. If the original position is x and y. Scaling factors are S_x and S_y then the value of coordinates after scaling will be x^1 and y_1.

If the picture to be enlarged to twice its original size then $S_x = S_y = 2$. If S_x and S_y are not equal then scaling will occur but it will elongate or distort the picture.

If scaling factors are less than one, then the size of the object will be reduced. If scaling factors are higher than one, then the size of the object will be enlarged.

If $S_x$and $S_y$are equal it is also called Uniform Scaling. If not equal then called Differential Scaling. If scaling factors with values less than one will move the object closer to coordinate origin, while a value higher than one will move coordinate position farther from origin.



$$x' = S_x \cdot x$$
$$y' = S_y \cdot y$$
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$
$$S_{S_x, S_y} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

Matrix for Scaling

3x3 Matrix

$$S = \begin{vmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

2x2 Matrix

$$\begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

Rotation:

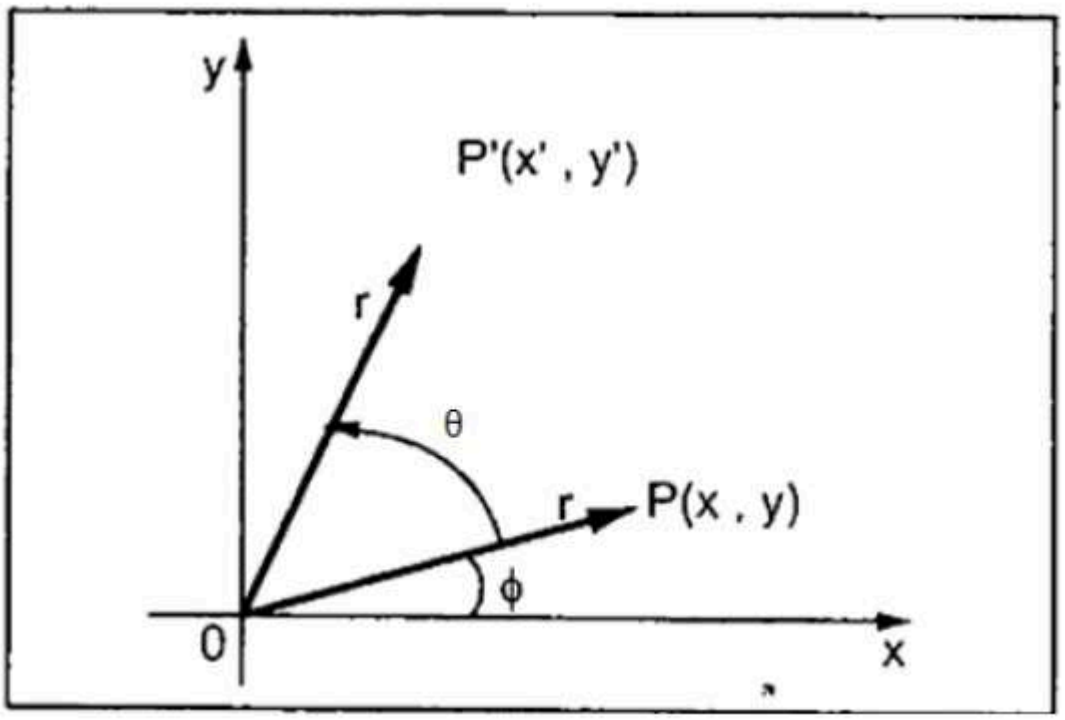
It is a process of changing the angle of the object. Rotation can be clockwise or anticlockwise. For rotation, we have to specify the angle of rotation and rotation point. Rotation point is also called a pivot point. It is printed about which object is rotated.

Types of Rotation:

1. Anticlockwise or counter-clockwise
2. clockwise

The positive value of the pivot point (rotation angle) rotates an object in a counter-clockwise (anti-clockwise) direction.

The negative value of the pivot point (rotation angle) rotates an object in a clockwise direction.



Using standard trigonometric the original coordinate of point PX, Y can be represented as –

$$X = r\cos\phi \ldots\ldots (1)$$

$$Y = r\sin\phi \ldots\ldots (2)$$

Same way we can represent the point P′ X', Y' as –

$$x' = r\cos(\phi + \theta) = r\cos\phi\cos\theta - r\sin\phi\sin\theta \ldots\ldots (3)$$

$$y' = r\sin(\phi + \theta) = r\cos\phi\sin\theta + r\sin\phi\cos\theta \ldots\ldots (4)$$

Substituting equation 1 & 2 in 3 & 4 respectively, we will get

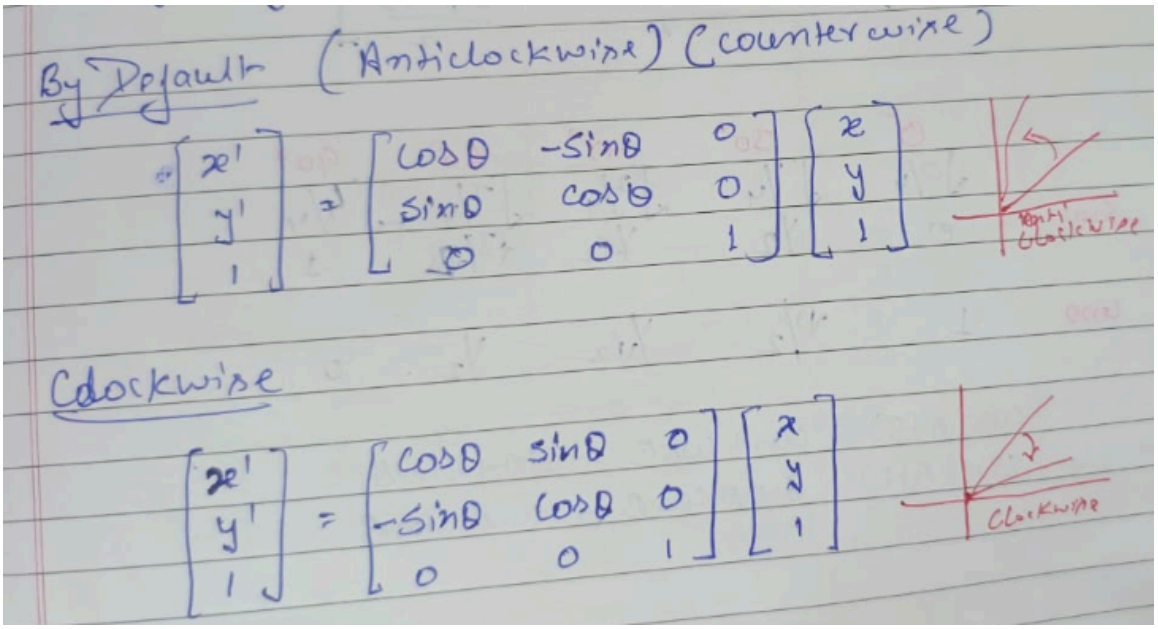
$$x' = x\cos\theta - y\sin\theta$$

$$y' = x\sin\theta + y\cos\theta$$

Representing the above equation in matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

3x3 Matrix



Reflection: It is a transformation which produces a mirror image of an object. The mirror image can be either about x-axis or y-axis. The object is rotated by180°.

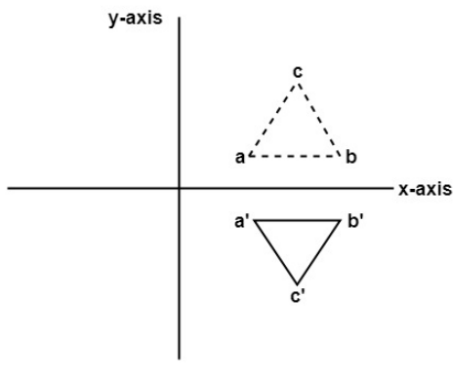
Types of Reflection:

1. Reflection about the x-axis
2. Reflection about the y-axis
3. Reflection about an axis perpendicular to xy plane and passing through the origin

1. **Reflection about x-axis:** The object can be reflected about x-axis with the help of the following matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In this transformation the value of x will remain the same whereas the value of y will become negative. Following figure shows the reflection of the object axis. The object will lie on the other side of the x-axis.

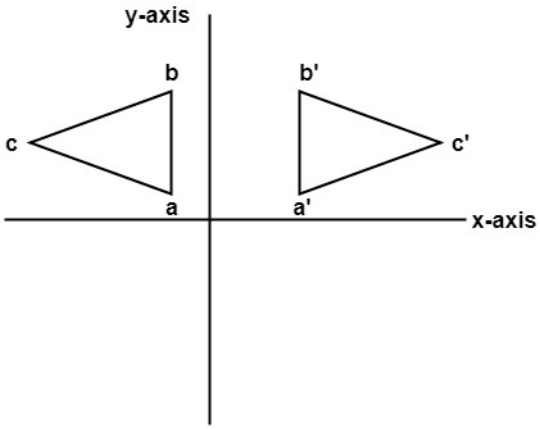


2. **Reflection about y-axis:** The object can be reflected about y-axis with the help of following transformation matrix

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

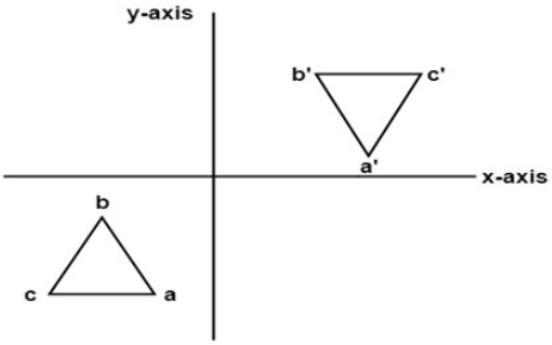
Here the values of x will be reversed, whereas the value of y will remain the same. The object will lie another side of the y-axis.

The following figure shows the reflection about the y-axis



3. **Reflection about an axis perpendicular to xy plane and passing through origin:**
   In the matrix of this transformation is given below

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

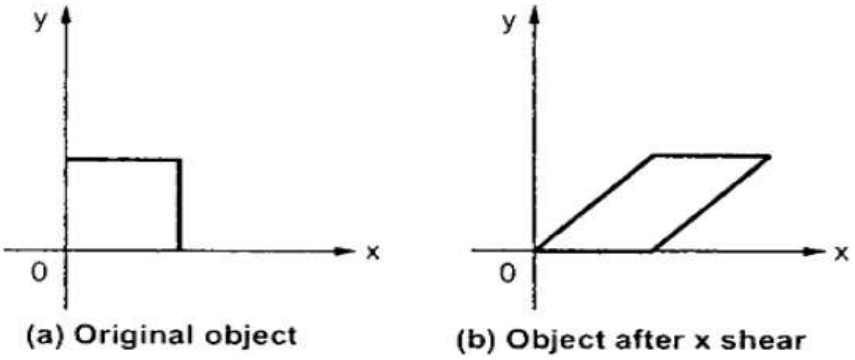


In this value of x and y both will be reversed. This is also called as half revolution about the origin.

Shearing: A transformation that slants the shape of an object is called the shear transformation. There are two shear transformations **X-Shear** and **Y-Shear**. One shifts X coordinates values and other shifts Y coordinate values. However; in both the cases only one coordinate changes its coordinates and other preserves its values. Shearing is also termed as **Skewing**.

X-Shear

The X-Shear preserves the Y coordinate and changes are made to X coordinates, which causes the vertical lines to tilt right or left as shown in below figure.



(a) Original object (b) Object after x shear

The transformation matrix for X-Shear can be represented as –

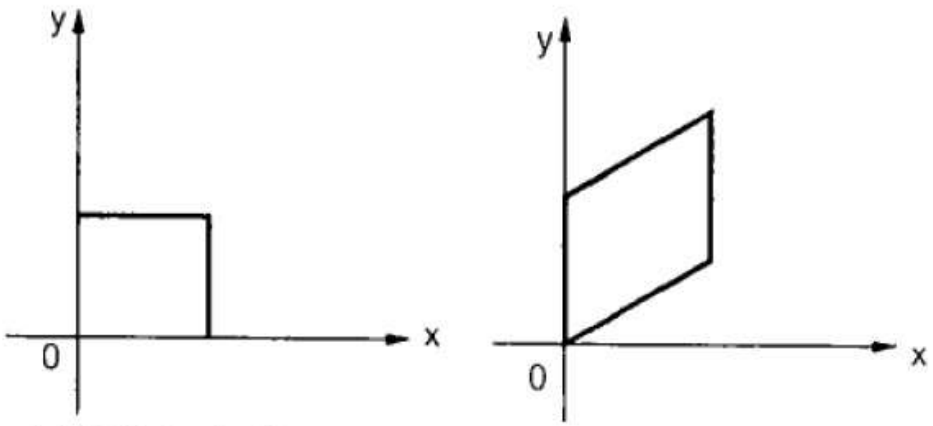
$$\mathbf{x' = x + sh_{x.}y}$$

$$\mathbf{y' = 0.x + y}$$

$$X_{sh} = \begin{bmatrix} 1 & shx & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Y-Shear

The Y-Shear preserves the X coordinates and changes the Y coordinates which causes the horizontal lines to transform into lines which slopes up or down as shown in the following figure.



(a) Original object (b) Object after y shear

The Y-Shear can be represented in matrix from as –

$$\mathbf{x'=x + 0.y}$$

$$\mathbf{y'=sh_{y}.x + y}$$

$$Y_{sh}\begin{bmatrix} 1 & 0 & 0 \\ shy & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shearing in X-Y directions: Here layers will be slided in both x as well as y direction. The sliding will be in horizontal as well as vertical direction. The shape of the object will be distorted. The matrix of shear in both directions is given by:

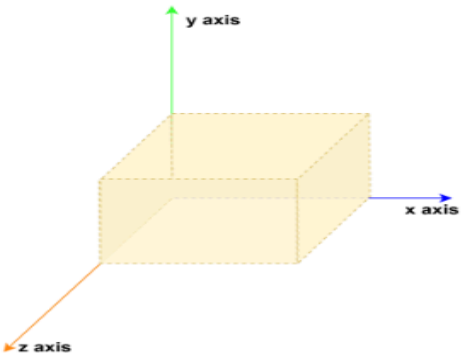
$$\begin{bmatrix} 1 & Sh_y & 0 \\ Sh_x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Composite Transformation:

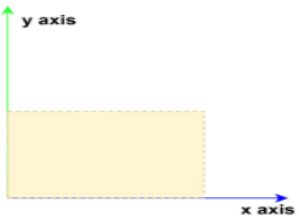
A number of transformations or sequences of transformations can be combined into a single one called composition. The resulting matrix is called a composite matrix. The process of combining is called concatenation.

Projection:

“Projection is a technique or process which is used to transform a 3D object into a 2D plane." In other words, we can define "projection as a mapping of points P (x, y, z) on to its image P' (x,' y,'z') in the projection plane or view plane, which create the display surface.”



3D Figure

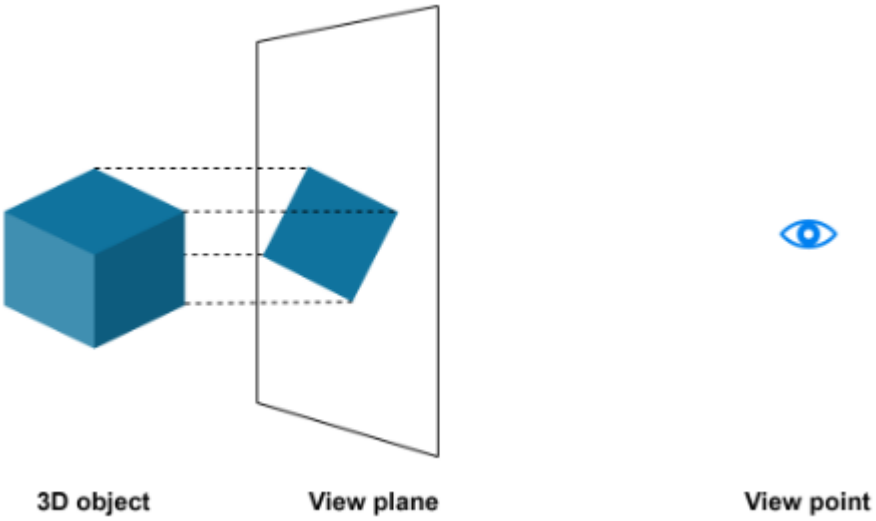


2D Figure

Parallel Projection:

In this type of projection, the object in the real world is projected onto a plane exactly parallel to the 3D object. Additionally, here the lines of projection remain a straight line.

Further, we can visualize parallel projection using an example:

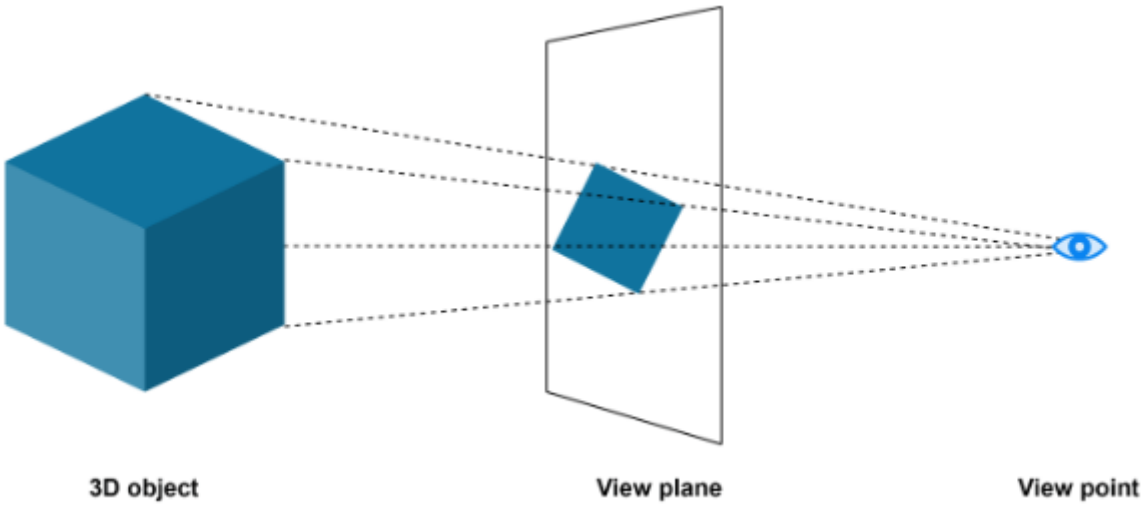


Perspective Projection

In the perspective projection, the distance of the project plane from the center of projection is finite. The object size keeps changing in reverse order with distance.

Perspective projection is used to determine the projector lines coming together at a single point. The single point is also called "project reference point" or "Center of projection."

Let's understand the perspective projection using an example:



Characteristic of Perspective Projection:

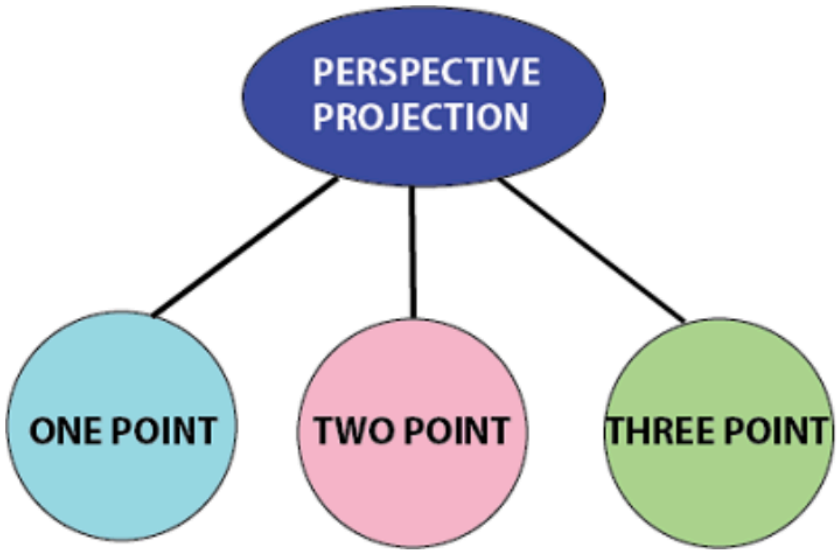
- The Distance between the object and projection center is finite.
- In Perspective Projection, it is difficult to define the actual size and shape of the object.
- The Perspective Projection has the concept of vanishing points.
- The Perspective Projection is realistic but tough to implement.

Vanishing Point: Vanishing point can be defined as a point in the image plane where all parallel lines are interlinked. The Vanishing point is also called “Directing Point.”

Use of Vanishing Point:

- It is used in 3D games and graphics editing.
- It is also used to represent 3D objects.
- We can also include perspective in the background of an image.
- We can also insert the shadow effect in an image.

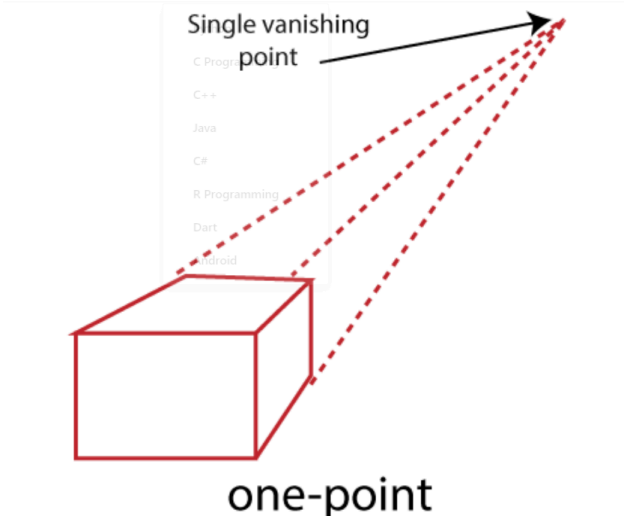
Types of Perspective Projection:



There are three types of Perspective Projection.

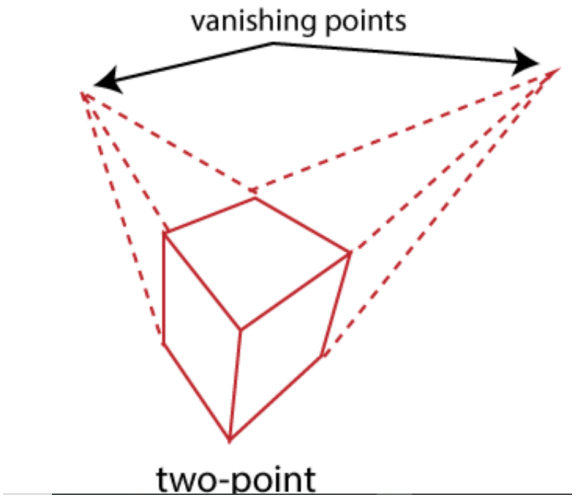
One Point: A One Point perspective contains only one vanishing point on the horizon line.

It is easy to draw.



Use of One Point- The One Point projection is mostly used to draw the images of roads, railway tracks, and buildings.

2.Two Point: It is also called "Angular Perspective." A Two Point perspective contains two vanishing points on the line.



Use of Two Point- The main use of Two Point projection is to draw the two corner roads.

3.Three-Point- The Three-Point Perspective contains three vanishing points. Two points lie on the horizon line, and one above or below the line.
It is very difficult to draw.
When we see an object from above, than the third point is below the ground. If we see an object from the below, than the third point is in the space above.

Use of Three-Point: It is mainly used in skyscraping.

Advantages:
Better Look
Clear Representation
Disadvantages:
Difficult to Draw
Not Suitable for many-dimensional images

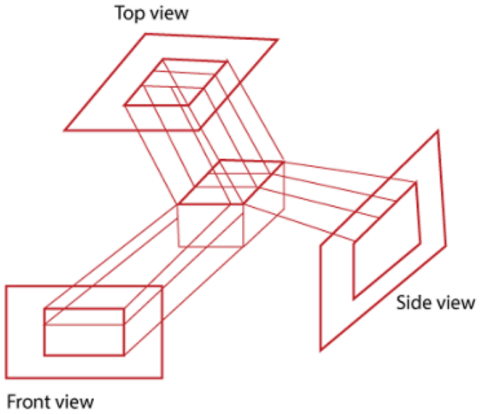
Parallel Projection:

In Parallel Projection, the distance of the project plane from the center of projection is infinite. We can specify the direction of projection instead of the center of the projection. We can connect the projected vertices through the line segment.
The parallel Projection eliminates the Z-Coordinate. And the parallel lines from each vertex in the object are enhanced until the lines intersect the view plane.

Characteristic of parallel Projection:
In parallel Projection, the projection lines are parallel to each other.
There is the least amount of distortion within the object.
The lines that are parallel to the object are also parallel to the drawing.
The view of Parallel Projection is less realistic because of no foreshortening.
The Parallel Projections are good for accurate measurements.

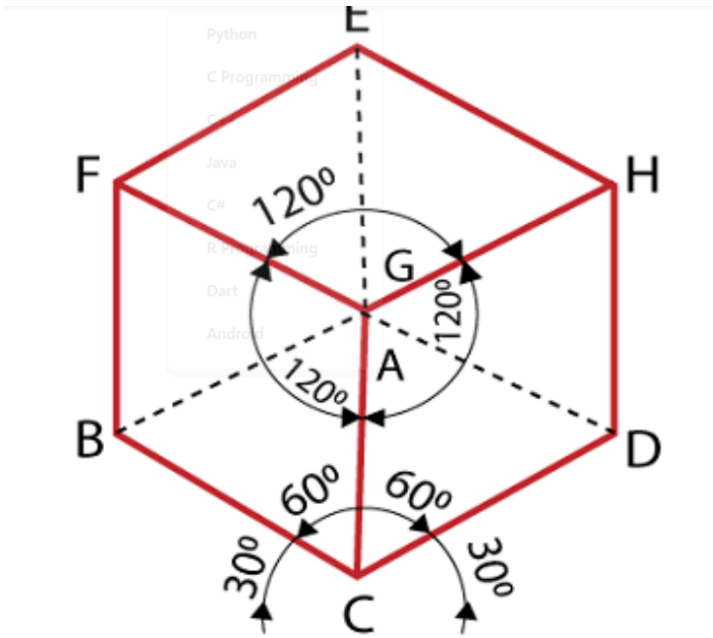
Types of Parallel Projection
There are two kind of parallel Projection:
Orthographic projection: In the Orthographic Parallel Projection, the Projection is perpendicular to the view plane.
The Orthographic Projection is divided into two parts-

Multiview Orthographic Projection: In Multiview Orthographic Projection, we can represent the two-dimensional Orthographic image into a three-dimensional object. The Multiview Orthographic Projection Includes-
Front View
Top View
Side View

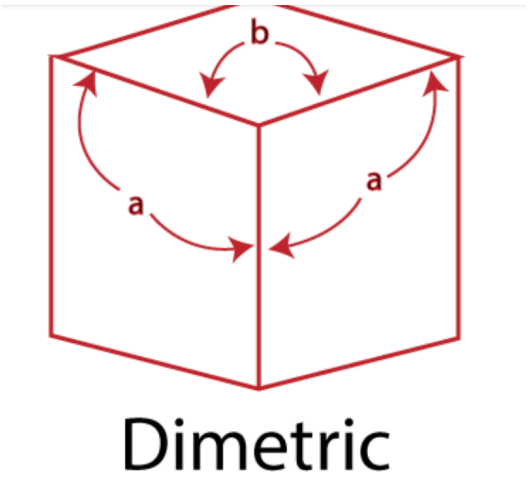


Axonometric Orthographic Projection: The Axonometric Orthographic Projection is used to construct the pi The Axonometric Orthographic Projectctorial representation of an object. The sight lines are perpendicular to the projection plane.

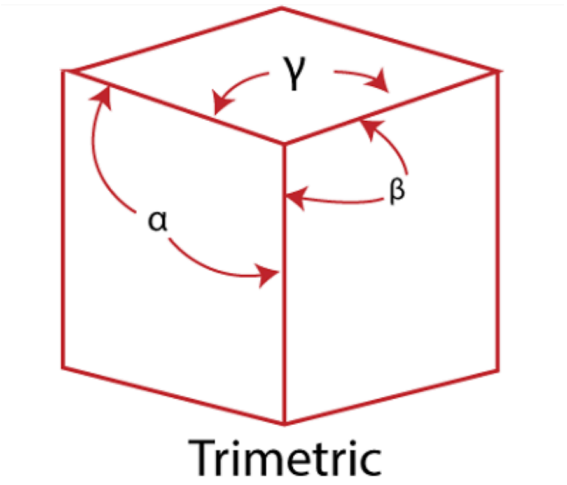
ion includes-
Isometric: In Isometric, we can represent the three-dimensional objects into two-dimensional drawings visually. The Angle between the two coordinate is 120 degrees.



2. Dimetric: In Dimetric Projection, the view direction of the two axes are equal, and the direction of the third axis is defined individually.



3. Trimetric: In the Trimetric Projection, the view direction of all three axes is unequal. The scale of all three angles is defined individually.

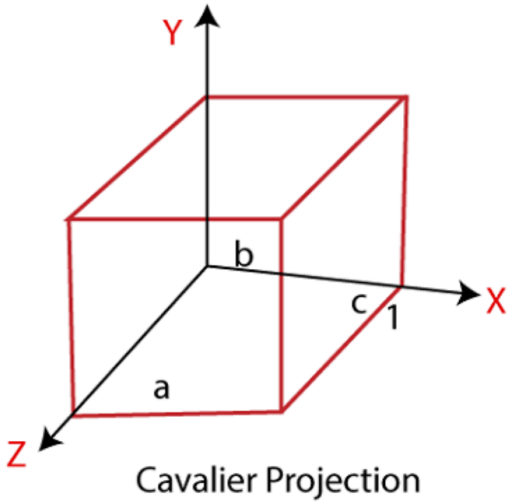


Oblique Projection: In the Oblique Parallel Projection, the direction of projection is not normal to projection of plane. It is a simple technique that is used to construct two-dimensional images of three-dimensional objects.

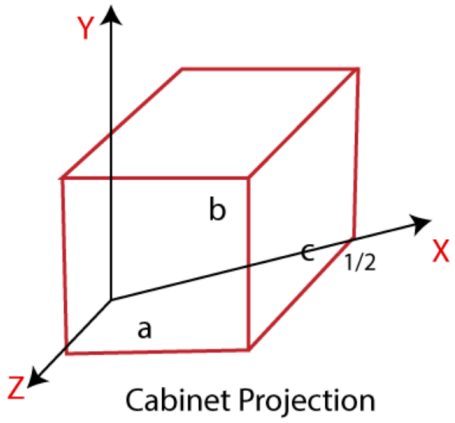
The Oblique Projection is mostly used in technical drawing.

The Oblique Projection is divided into two parts-

Cavalier: In cavalier Projection, there is an angle between the Projection and Projection Plane is 45 degrees



- **Cabinet: In Cabinet Projection, there is an angle between Projection and Projection Plane is 63.4 degrees.**



Advantages:

1. **Good for exact Measurement**
2. **Parallel lines remain parallel**

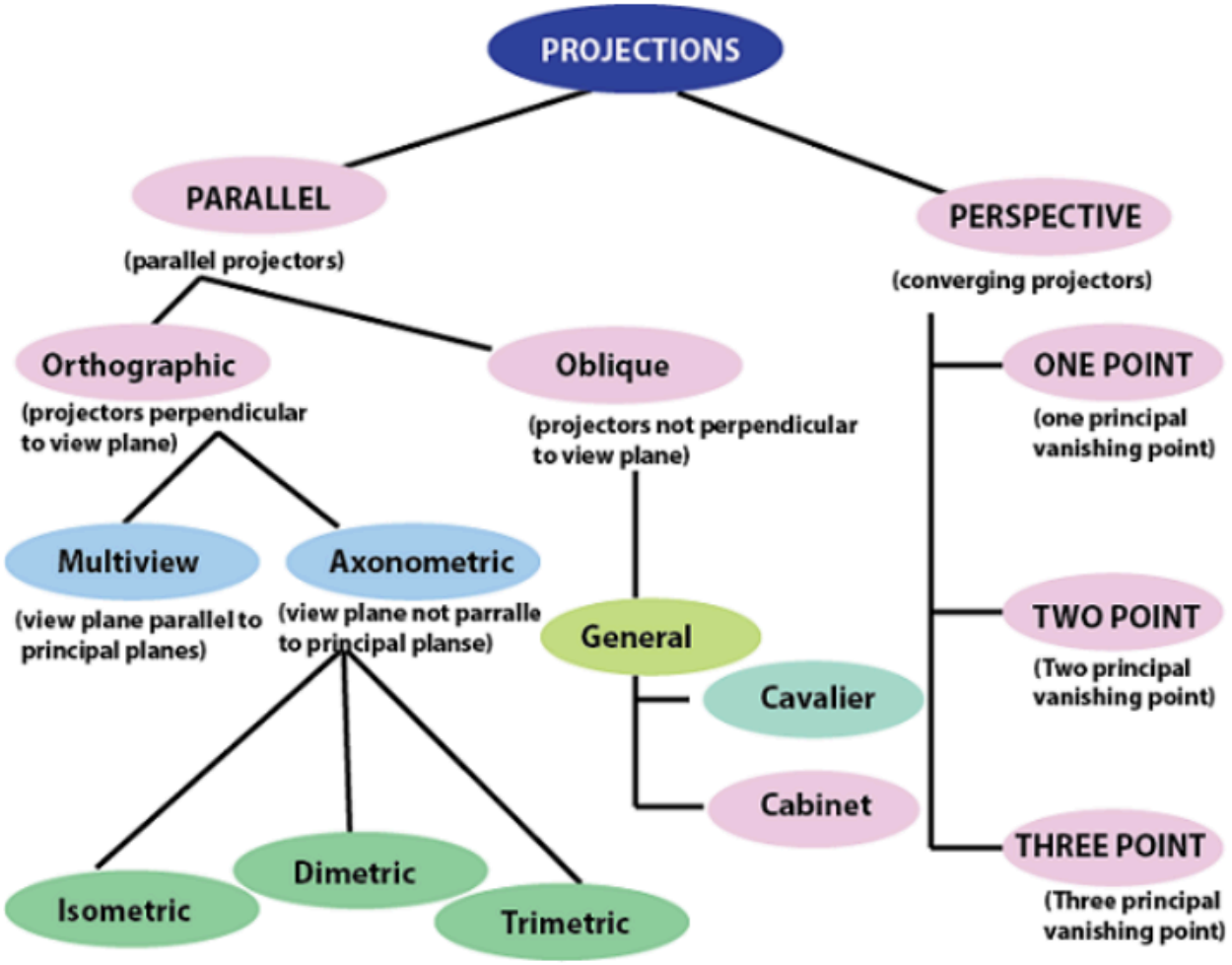
Disadvantages:

1. **Less Realistic Looking**
2. **Angles are not preserved**

Comparison of Parallel and Perspective Projections

Parallel Projection	Perspective Projection
Size of 3D figure and 2D figure remains same	Size of 3D figure is larger than 2D figure
Shape of 3D figure and 2D figure remains same	Shape of 3D figure and 2D figure differs based on the viewpoint location as well as angle
Lines of projection are straight	Lines of projection converges to a view-point
The distance between view plane and viewpoint is infinite	Distance between view plane and view-point is finite since it converges into the view point
Actual size of the 3D object is known	Actual size of the 3D object is unknown
An example is X-ray of teeth	An example is a picture of mountain taken using camera

Taxonomy of Projections



Center of Projection:

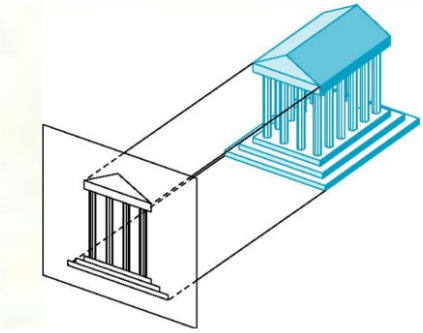
It is an arbitrary point from where the lines are drawn on each point of an object.

1. If cop is located at a finite point in 3D space , Perspective projection is the result
2. If the cop is located at infinity, all the lines are parallel and the result is a parallel projection.

Parallel Projection

1. Orthographic Projection:

Projectors are orthogonal to projection surface



Orthographic projection is a means of representing three-dimensional objects in two dimensions. It is a form of parallel projection, in which all the projection lines are orthogonal to the projection plane.

Preserves both distances and angles

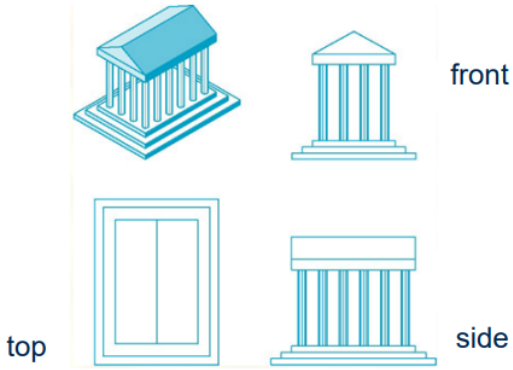
– Shapes preserved

– Can be used for measurements

- Building plans
- Manuals

Multiview Orthographic Projection

- Projection plane parallel to principal face Usually form front, top, side views



Window Coordinate System

Window: “The process of selecting and viewing an image with different views, called windowing.”

All the objects in the real world have a size. We can measure the size and location of an object by the unit.

For Example-We use the meter unit to measure both size and the location of the object.

When we represent the image of an object on the screen, then we use the screen coordinate system. The screen coordinate system is used to define the location of the object. When we select the screen coordinate system, then the image can be displayed on the screen.

“The Capability to show some part of an object in a window is known as windowing.”

“The rectangular area describes in the world coordinate system is called the window.”

Viewport:

“The viewport can be defined as an area on the screen which is used to display the object.” The window is an area space for the object. Viewport surrounds the object. We need the coordinate transformation to display the object on the screen. We can define many viewports on a different area of the screen and also see the same object from a different angle in the viewport.

Viewing Transformation

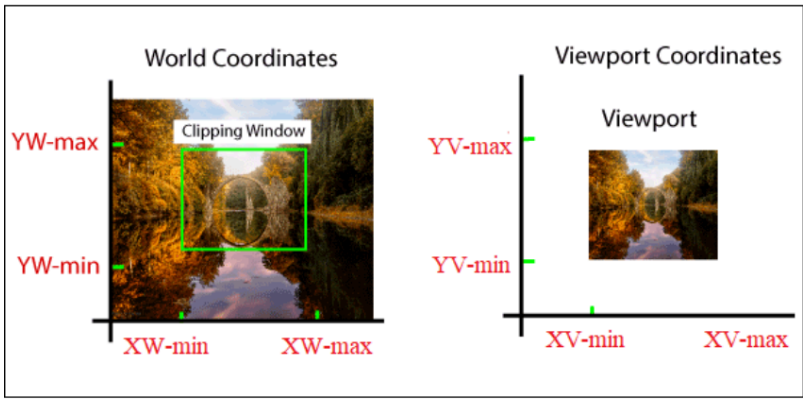
Window to Viewport transformation:

“Window to viewport transformation is a process of converting two-dimensional or world into a device coordinate.”

The object inside the clipping window is mapped to the viewport. The viewport is displayed inside the interface window on the screen. We can use the clipping window to select the part of an object, and the viewport is used to display the selected part of the object on the screen or output device.

“The process of selecting and viewing an image with different views, called windowing.”

All the objects in the real world have a size. We can measure the size and location of an object by the unit.



Steps for Window to Viewport Transformation: We can follow the following steps for transform window to viewport:

Step 1: Translation of the window towards the Origin- If we shift the window towards the origin, the upper left, and the lower-left corner of the window will be negative (-). The translation factor also should be negative (-).

Step 2: Resize the window to viewport size- To Convert the size of the window into the size of the viewport, we will use the following formulas:

Sx = XVmax – XVmin / XWmax – XWmin

Sy = YVmax – YVmin / YWmax – YWmin

Step 3: Translation of window (Position of window and viewport must be same)- If the lower-left corner of viewport is (0, 0), then the window lower-left corner is already shifted on origin after following step 1.

If the lower-left corner is not equal to (0, 0), then the translation factor should be positive (+).

It may be possible that sometimes the size of the viewport is greater or smaller than the window. In that situation, we have to enlarge or compress the size of the window according to the viewport. We can perform it using some mathematical calculations.

A point on window = (XW, YW)

Corresponding point on viewport = (XV, YV)

To calculate (XV, YV) –

Normalized point on window = (XW- XWmin / XWmax - XWmin)

(YW- YWmin / YWmax - YWmin)

Normalized point on viewport = (XV- XVmin / XVmax - XVmin)

(YV- YVmin / YVmax - YVmin)

Now, the position of object in viewport and window are same-

For Coordinate x:

(XW- XWmin / XWmax - XWmin) = (XV- XVmin / XVmax - XVmin)

For Coordinate y:

(YW- YWmin / YWmax - YWmin) = (YV- YVmin / YVmax - YVmin)

Now, we get

XV = XVmin + (XW- XWmin) Sx

YV = YVmin + (YW- YWmin) Sy

Here Sx and Sy are the scaling factor for x and y coordinate.

$$Sx = XVmax - XVmin / XWmax - XWmin$$

$$Sy = YVmax - YVmin / YWmax - YWmin$$